

IFIT1 is rapidly evolving and exhibits disparate antiviral activities across 11 mammalian orders

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Abstract

Mammalian mRNAs possess an N7-methylguanosine (m7G) cap and 2'O methylation of the initiating nucleotide at their 5' end, whereas certain viral RNAs lack these characteristic features. The human antiviral restriction factor IFIT1 recognizes and binds to specific viral RNAs that lack the 5' features of host mRNAs, resulting in targeted suppression of viral RNA translation. This interaction imposes significant host-driven evolutionary pressures on viruses, and many viruses have evolved mechanisms to evade the antiviral action of human IFIT1. However, little is known about the virus-driven pressures that may have shaped the antiviral activity of IFIT1 genes across mammals. Here, we take an evolution-guided approach to show that the IFIT1 gene is rapidly evolving in multiple mammalian clades, with positive selection acting upon several residues in distinct regions of the protein. In functional assays with 39 IFIT1s spanning diverse mammals, we demonstrate that IFIT1 exhibits a range of antiviral phenotypes, with many orthologs lacking antiviral activity against viruses that are strongly suppressed by other IFIT1s. We further show that IFIT1s from human and a bat, the black flying fox, inhibit Venezuelan equine encephalitis virus (VEEV) and strongly bind to Cap0 RNAs. Unexpectedly, chimpanzee IFIT1, which differs from human IFIT1 by only 8 amino acids, does not inhibit VEEV infection and exhibits minimal Cap0 RNA-binding. In mutagenesis studies, we determine that amino acids 364 and 366, with the latter undergoing positive selection, are sufficient to confer the differential anti-VEEV activity between human and chimpanzee IFIT1. These data suggest that virus-host genetic conflicts have influenced the antiviral specificity of IFIT1 across diverse mammalian orders.

eLife Assessment

This **valuable** report describes the changing antiviral activity of IFIT1 across mammals and in response to distinct viruses, likely as a result of past arms races. One of the main strengths of the manuscript is the breadth of mammalian IFIT1 orthologs and viruses that were tested. Overall the evidence is **solid**, but the analysis of positive selection could benefit from more thorough validation with complementary selection tests and also from assessing or more extended discussion of the impact of recombination and/or physical interactions with other IFITs.

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Introduction

Interferon-induced protein with tetratricopeptide repeats 1 (IFIT1) is an interferon-stimulated gene (ISG) that is capable of distinguishing between self and non-self RNA. Mammalian mRNAs possess an N7-methylguanosine (m7G) cap and 2'O methylation of the initiating nucleotide at their 5' end. Many viral RNAs, however, lack these characteristic features. IFIT1 is able to recognize and bind viral RNAs that lack a m7G cap, including 5' triphosphate (5'-ppp-RNA) or viral RNAs that possess a m7G cap but lack 2'O methylation of the initiating nucleotide (Cap0 RNA) (1, 2). Crystal structures of human IFIT1 bound to both 5'-ppp-RNA and Cap0 RNA have helped uncover biochemical and structural determinants of IFIT1 antiviral activity, including residues that confer RNA-binding properties (3). IFIT1 binding to viral RNA subsequently inhibits infection by decreasing the amount of accessible RNA from the pool of replicating viral RNA (4) and by interacting with eIF proteins required for translation of viral proteins (5).

Viral restriction via IFIT1 binding to non-self RNAs imposes a host-driven evolutionary pressure for viruses to evade the antiviral function of IFIT1, as many viruses have evolved various mechanisms to elude IFIT1 binding and viral suppression (5). This includes viruses encoding their own capping and 2'O methyltransferase enzymes, using internal ribosome entry sites (IRES) to allow cap-independent translation of viral proteins, using host mRNA caps through cap-snatching, or hindering IFIT1 binding with RNA structures (5). For the alphavirus Venezuelan equine encephalitis virus (VEEV), which has a positive-sense, single-stranded RNA (+ssRNA) genome, a secondary RNA structure in the 5' UTR of the pathogenic Trinidad strain occludes IFIT1 binding to viral RNA, whereas the IFIT1-sensitive TC-83 strain has a point mutation that destabilizes this RNA structure. This allows IFIT1 to bind viral RNA and inhibit viral infection (1). VEEV TC83 can be inhibited by both human IFIT1 and mouse Ifit1B, suggesting similar mechanisms have been at least partially evolutionarily conserved to restrict viral infection by IFIT proteins. However, as viruses have evolved mechanisms to evade IFIT1 antiviral activity, IFIT1 may also experience evolutionary pressure to overcome viral evasion. We know little about how such virus-imposed constraints may have influenced *IFIT1* evolution across mammals, broadly.

Pathogenic viruses can cause severe disease or death in mammals, imposing a major selective pressure on their mammalian hosts to evolve mechanisms that restrict viral infection. In turn, there is pressure on viruses to evade host defenses. These “molecular arms races” of host-pathogen co-evolution are evident as many genes involved in antiviral immunity exhibit signatures of positive selection or rapid evolution. Computational evolutionary analyses of antiviral ISGs have revealed specific domains and residues that are rapidly evolving (6–8). Identifying these signatures of positive selection has uncovered functional determinants of antiviral function in antiviral effectors such as MX1 and TRIM5α (9, 10). In addition to evolutionary pressure from host-pathogen molecular arms races, IFIT1 may have also been influenced by gene recombination occurring between *IFIT1* and the *IFIT1B* gene (11). *IFIT1* genes have undergone duplication, gene loss, and recombination with *IFIT1B* in mammals, resulting in genes that exhibit differential antiviral activity. However, this was only assessed in a few species, and the extent of differential antiviral activity of IFIT1 across Mammalia remains unknown. Recently, in a screen for antiviral ISGs from bats, we found that IFIT1 from the black flying fox displays particularly strong suppression of VEEV (12).

Together, these studies led us to hypothesize that evolution may have resulted in “species-specific” antiviral ability of IFIT1 orthologs. While previous studies have provided considerable insight into the antiviral mechanism and structure of some mammalian IFIT1 proteins, a comprehensive study into the antiviral specificity of IFIT1 orthologs remains lacking. Accordingly, we sought to use evolutionary and functional methods to interrogate differences in antiviral activity and biochemical properties of mammalian IFIT1.

Results

IFIT1 is rapidly evolving in mammals

Many antiviral effectors exhibit signatures of positive selection, as restriction factors and the viruses they suppress are often engaged in “molecular arms races” (6, 7). Therefore, we hypothesized IFIT1 may be rapidly evolving, as IFIT1 is a potent antiviral effector that many viruses have evolved mechanisms to evade (5). To test this, we used established computational methods to determine if the *IFIT1* gene is rapidly evolving across diverse clades of mammals. Using Phylogenetic Analysis using Maximum Likelihood (PAML) (13), we determined that *IFIT1* exhibits signatures of rapid evolution in all clades tested (primates, bats, carnivores, and ungulates) (Fig 1A). Rodents were not included in this analysis, as a genetic ortholog of IFIT1 has been lost in many rodent species (11). While all lineages showed signs of rapid evolution, carnivore IFIT1s exhibited the least significant likelihood of pervasive positive selection. However, all mammalian lineages were also determined to have at least 2 sites of statistically significant positive selection (Fig 1A). Detection of positive selection in certain domains or at specific sites of restriction factors has been utilized to uncover genetic determinants of antiviral function (6, 7). Therefore, we sought to potentially uncover other sites in IFIT1 that may be engaged in host-pathogen conflicts using complementary methods to PAML. We next used Fast Unconstrained Bayesian Approximation (FUBAR) (14), another approach that had been used to identify rapidly evolving sites of restriction factors. Using FUBAR analysis, we further identified at least one site predicted to be undergoing significant rapid evolution for all clades except carnivores (Fig 1B). Another complementary analysis, Mixed Effects Model of Evolution (MEME) (15), identified several sites of positive selection for all lineages of tested mammals (Fig 1C). To our knowledge, these signatures of positive selection in *IFIT1* have not previously been reported and suggest that evolutionary pressure has acted on multiple regions of IFIT1.

Human IFIT1 residue 193 exhibits mutational resiliency

We next focused on three codons in primate IFIT1 – encoding V170, L193, and L366 in the human ortholog – that were determined by at least two models to be evolving under positive selection (Fig 1D). We examined the location of each in solved IFIT1 structures for potential importance in RNA-binding. None of these residues had previously been reported to be involved in IFIT1 RNA-binding or antiviral function (2, 3). While crystal structures of human IFIT1 bound to RNA indicate that residues V170 and L366 are not near the RNA-binding region, L193 is part of the TPR4 loop which has been reported to act as a lid over the 3' exit of the RNA-binding tunnel (3) (Fig 2A, 2B). The TPR4 loop, located between α -helices 9 and 10, also has been suggested to play a role in mediating contacts between subdomain II and subdomain III (Fig 2B) (3). Specifically, L193 in human IFIT1 protrudes into the exit of the RNA-binding tunnel and comes into close contact with RNA (Fig 2C) (3). In addition, L193 is in close proximity to R187, which is required for IFIT1 RNA binding and antiviral function (2). Therefore, we sought to test the importance of L193 in the antiviral function of IFIT1. Our general strategy was to use CRISPR-Cas9 gene editing to silence endogenous IFIT1 in human cells, and then reconstitute expression by ectopic expression of IFIT1 193 mutants. We performed site-directed mutagenesis to generate point mutations in a CRISPR-resistant human IFIT1 clone that correspond to every possible amino acid at residue 193. Huh7.5 cells that were CRISPR-targeted for endogenous IFIT1 were transduced with lentiviral pseudoparticles to express each of the 193 mutant IFIT1s or appropriate controls. Cells expressing the IFIT1 point mutants were then challenged with VEEV encoding a GFP reporter, and infection was quantified by flow cytometry (Fig 2D). We observed that most amino acid substitutions at residue 193 were tolerated by human IFIT1 with respect to antiviral function. In addition, western blotting determined that all IFIT1 point mutants were expressed to appreciable levels (Fig 2D, Bottom). Indeed, all mutants at position 193, whether hydrophobic or

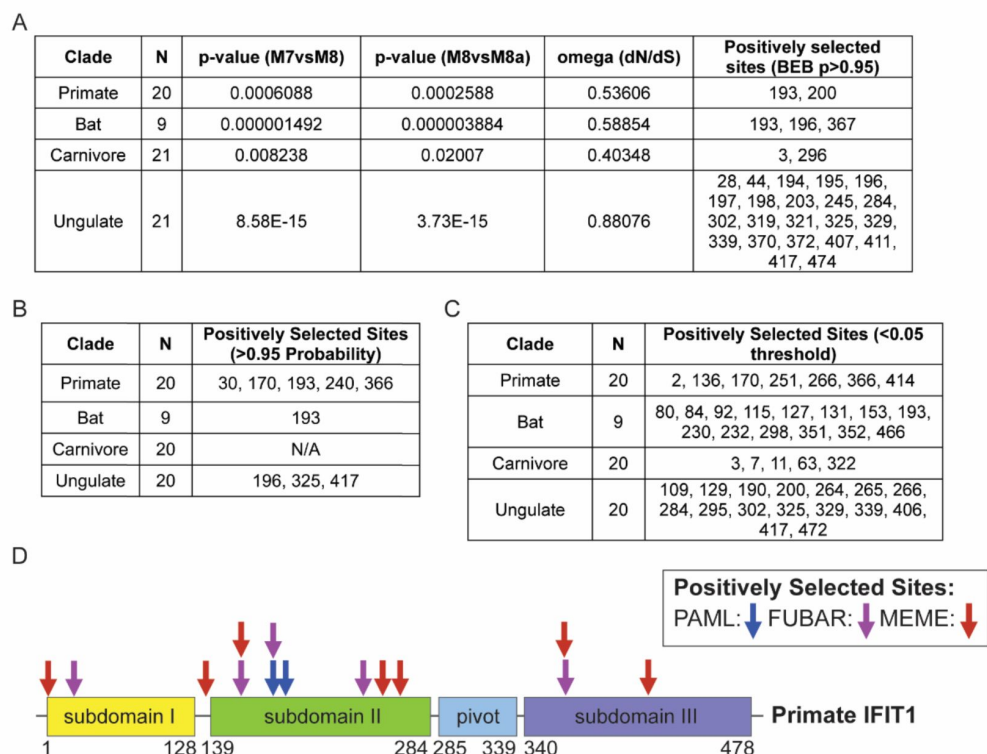


Figure 1.

Evolutionary analysis of IFIT1 from clades of mammals reveals rapid evolution.

A. PAML analysis on NCBI-derived coding sequences that were trimmed aligned using CodeML. Likelihood ratio tests were performed to compare model 7 vs model 8 and model 8 vs model 8a to determine the presence of positive selection. N=number of sequences input, and numbers under "positively selected sites" represent residue number of the reference sequence. B. FUBAR analysis on coding sequences that were trimmed and aligned using HyPhy software and DataMonkey. N=number of sequences input, and numbers under "positively selected sites" represent residue number of the reference sequence. C. MEME analysis on coding sequences that were trimmed and aligned using HyPhy software and the DataMonkey application. N=number of sequences input, and numbers under "sites of episodic positive selection" represent residue number of the reference sequence. D. Diagram of Primate IFIT1 domains and location of positively selected sites determined by PAML, FUBAR, and MEME.

positively charged, inhibited VEEV similarly to the wild-type (193L) IFIT1 (**Fig 2D**). Polar amino acids introduced at position 193 also retained antiviral function (**Fig 2D**). Severe loss of function in anti-VEEV activity of IFIT1 was only notable with glycine or proline (~50% loss of antiviral function) or with negatively charged residues (~75-100% loss of antiviral function) (**Fig 2D**). This experiment demonstrated the resiliency of IFIT1 antiviral function despite introduction of mutations in rapidly evolving site 193 within the TPR4 loop.

IFIT1 orthologs from diverse mammals demonstrate species-specific antiviral activity

To better understand the consequences of the evolutionary pressures on IFIT1, we sought to test the antiviral activity of diverse mammalian IFIT1 orthologs against RNA viruses. To accomplish this, we cloned IFIT1 protein-coding sequences from 38 species (including 2 predicted IFIT1 sequences from horse) of mammals into lentiviral expression vectors. The chosen orthologs ranged from 60% to over 98% identical to human IFIT1 in protein coding sequence and encompassed 11 taxonomic orders (**Fig 3A**, **Fig 3B**, Supp Table 1) within the superorders Euarchontoglires, Laurasiatheria, Afrotheria, and Xenartha, thus representing extensive diversity in both sequence and evolutionary distance relative to human IFIT1. All IFIT1 orthologs were also N-terminally HA-tagged to allow for detection by western blot. After generating lentivirus pseudoparticles, we transduced human Huh7.5 cells with these 39 IFIT1 orthologs or appropriate controls and challenged IFIT1-expressing cells with two different RNA viruses encoding a GFP reporter, VEEV and vesicular stomatitis virus (VSV). These viruses were chosen because of reported suppression by human IFIT1 through distinct mechanisms. VSV is a rhabdovirus with a negative sense single-stranded RNA genome. IFIT1 inhibits VSV by binding to viral RNA ending in a triphosphate (5'-ppp-N), sequestering viral RNA from the replicating RNA pool (2). VEEV (TC-83 strain) is an alphavirus with a positive-sense single-stranded RNA genome and is restricted by IFIT1 due to the binding and translation inhibition of viral Cap0 RNA (m7G-ppp-N). This viral RNA possesses the canonical m7G cap but lacks 2'O methylation of the initialing nucleotide of the viral 5' UTR, allowing recognition and binding by human IFIT1 (1, 16). IFIT1-expressing cells were challenged with an MOI of 1 for both VSV-GFP and VEEV-GFP, and infection was quantified by flow cytometry (**Fig 3C**). IFIT1 orthologs capable of potentially restricting VSV or VEEV infection, defined by inhibition of 70% or greater relative to control cells, were considered hits in this screen. VEEV infection was potentially suppressed by eleven IFIT1 orthologs (African savannah elephant, Chinese pangolin, sheep, white-tailed deer, black flying fox, dromedary camel, orca, Angola colobus, big brown bat, human, and large flying fox) (**Fig 3C**). Five IFIT1 orthologs (orca, cape elephant shrew, dolphin, European rabbit, and nine-banded armadillo) inhibited VSV infection by at least 70% compared to control cells (**Fig 3D**). We were surprised that hits for each virus spanned such diverse mammalian clades, and that there was little overlap in orthologs that inhibit VEEV and VSV. Also unexpected, many IFIT1 orthologs exhibited no antiviral activity against these two viruses, as only 10 of the 39 tested orthologs met our threshold to be considered hits. Western blotting determined that IFIT1 expression was not uniform during the ortholog screen (**Supp Fig 1**), suggesting that low expression of certain orthologs may have contributed to lack of antiviral function. Expression of certain IFIT1 orthologs (e.g. orca and European rabbit) was much higher than others, and some IFIT1 orthologs were expressed at very low levels (e.g. African green monkey and greater horseshoe bat) (**Supp Fig 1**). In addition, black-capped squirrel monkey IFIT1 was observed to run at a lower-than-expected molecular weight, suggesting a truncation or cleavage of the protein product (**Supp Fig 1**). However, it was noted that many IFIT1 orthologs were hits in the screens despite low expression, and only orca IFIT1 was a hit in both VSV and VEEV screens despite very high expression by multiple orthologs. These data suggest that IFIT1 orthologs may exhibit species-specific antiviral functions independent of differences in protein expression.

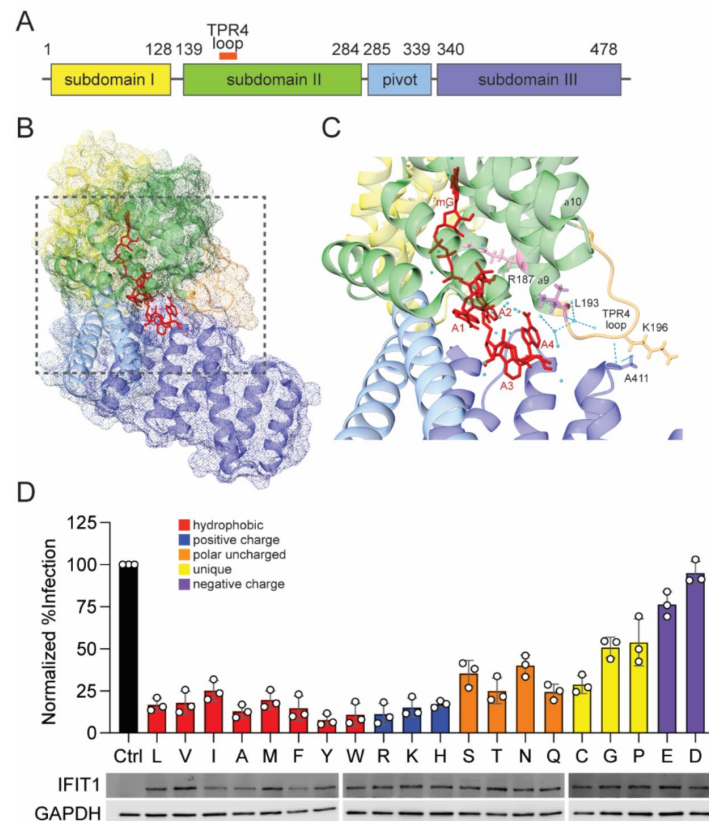


Figure 2.

Mutagenesis of rapidly evolving residue 193 in human IFIT1 reveals mutational resiliency.

A. Depiction of the human IFIT1 protein with the TPR4 loop location (orange) shown. B. Solved crystal structure of IFIT1 bound to RNA with TPR4 loop forming a “lid” over the exit of the RNA-binding tunnel (PDB:5udj). C. Zoom in on the TPR4 loop and RNA-binding tunnel exit illustrating the location of the TPR4 loop and residue 193 in relation to bound RNA in the human IFIT1 crystal structure. D. (Top) Saturating mutagenesis screen in cells expressing human IFIT1 with residue 193 mutated to every possible residue and challenged with VEEV infection. (Top) (Bottom) Western blot from lysates of Huh7.5 cells expressing IFIT1 point mutants.

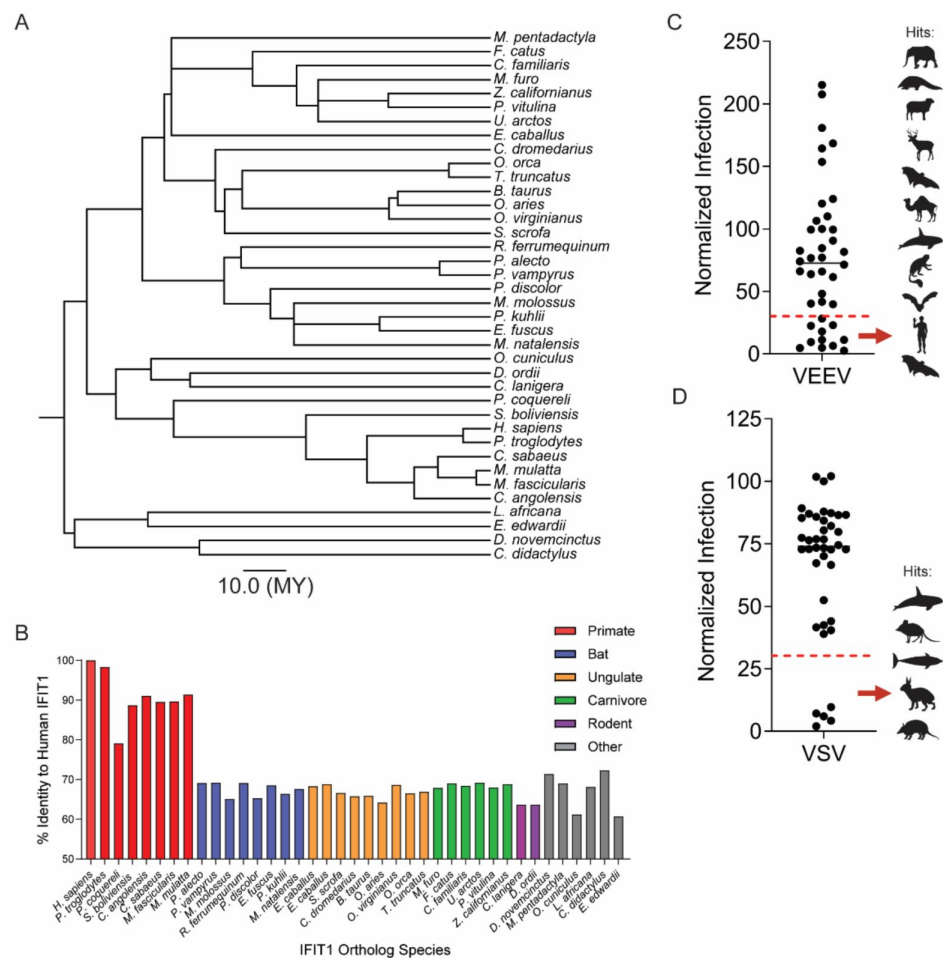


Figure 3.

IFIT1 ortholog screens reveal extensive heterogeneity in mammalian IFIT1 antiviral function.

A. TimeTree illustrating the evolutionary relationship of IFIT1 orthologs selected for ortholog screen. Scale bar represents divergence time of 10 million years. B. Graph of protein sequence identity of IFIT1 orthologs used in screen relative to human IFIT1. C. Dot plot representing the relative infection (compared to control cells) from an ectopic overexpression screen in which Huh7.5 cells expressing 39 different IFIT1 mammalian orthologs were challenged with 1.0 MOI VEEV-GFP for 4 h. Infectivity was quantified by flow cytometry. Red line denotes 30% relative infection. n=2 biological replicates. D. Same as C, for VSV-GFP (1 MOI, 4 h infection). n=3 biological replicates.

Since expression of IFIT1 from distinct species was not equal during the ortholog screen, we sought to validate the antiviral phenotypes of a subset of orthologs with more comparable expression. We selected 9 orthologs for follow-up based on protein expression, evolutionary diversity, and variable phenotypes exhibited in the ortholog screen. Those chosen for validation and follow-up included IFIT1 from Chinese pangolin (hereafter referred to as pangolin), orca, sheep, black flying fox, big brown bat, chinchilla, human, chimpanzee, and cape elephant shrew (hereafter referred to as shrew) (**Fig 4A**). Human Huh7.5 cells were transduced with differing doses of lentivirus pseudoparticles to normalize IFIT1 expression between this subset of orthologs. After puromycin selection, western blot analysis revealed that the expression of IFIT1 orthologs was relatively uniform, except for the higher expression of orca IFIT1 and notably lower expression of pangolin IFIT1 (**Fig 4B**). IFIT1-expressing cells were then challenged with VEEV (**Fig 4C**) and VSV (**Fig 4D**) at an MOI of 2 to validate results from the original IFIT1 ortholog overexpression screen. VSV was potently inhibited by orca and shrew IFIT1, while no other tested orthologs significantly suppressed infection. Thus, VSV may evade suppression by many mammalian IFIT1s. VEEV was potently inhibited by pangolin, sheep, black flying fox, big brown bat, and human IFIT1, while orca IFIT1 modestly inhibited VEEV. Strikingly, VEEV was not significantly inhibited by chimpanzee IFIT1 but was largely suppressed by human IFIT1. This was of keen interest as human and chimpanzee IFIT1 differ in protein sequence by only 8 amino acids, and one of these (residue 366) was identified as rapidly evolving by both FUBAR and MEME (**Fig 1D**). Together, these validation experiments demonstrate that mammalian IFIT1 species-specific differences in viral suppression are largely independent of expression differences. This is exemplified by potent restriction of VEEV by pangolin and big brown bat IFIT1 despite low expression (**Fig 4B** and **4C**). Furthermore, high-expressing sheep IFIT1 only restricted VEEV infection, but not VSV, and orca IFIT1 only modestly reduced VEEV infection despite much higher expression than all other IFIT1s (**Fig 4B-D**).

We further tested the ability of this subset of 9 IFIT1 orthologs to inhibit human parainfluenza virus type 3 (PIV3) (**Fig 4E**). PIV3 is a paramyxovirus with a negative sense, single-stranded RNA (–ssRNA) genome that is not previously reported to be inhibited by IFIT1. However, PIV3 was chosen for follow-up because human IFIT1 is a dominant antiviral effector towards the related human parainfluenza virus type 5 (PIV5) acting through an unknown mechanism (17). PIV3 was significantly restricted by shrew, pangolin, chinchilla, black flying fox, and big brown bat IFIT1 (**Fig 4E**). This was notable as human IFIT1 exhibited no antiviral activity towards PIV3, whereas it is a key component of IFN-mediated restriction of the related PIV5 (17). Chinchilla IFIT1 also exhibited disparate activities, as it did not inhibit VEEV (**Fig 4C**) or VSV (**Fig 4D**), yet significantly protected cells from PIV3 infection (**Fig 4E**). We then tested cells expressing the 9 selected IFIT1 orthologs for the ability to suppress another alphavirus related to VEEV, Sindbis virus (SINV) (**Fig 4E**). SINV was largely resistant to most mammalian IFIT1s, consistent with previous human IFIT1 studies (1, 16). However, SINV was potently suppressed by orca IFIT1, and modestly suppressed by pangolin, sheep, and chinchilla IFIT1 (**Fig 4E**), highlighting a pattern of suppression that was distinct across the 9 orthologs relative to the alphavirus VEEV. Together, these data demonstrate that IFIT1 orthologs exhibit broad antiviral phenotypes towards a panel of viruses with both +ssRNA and –ssRNA genomes (**Fig 4G**), largely independent of differences in protein expression.

Human, black flying fox, and chimpanzee IFIT1 protein exhibit species-specific RNA binding

As previously highlighted, human and black flying fox IFIT1 significantly protected cells from VEEV infection compared to a control, while chimpanzee IFIT1 did not (**Fig 4C**). The black flying fox (*Pteropus alecto*), is a model megabat species that is a reservoir for zoonotic viruses (18–20) and expresses restriction factors with unique antiviral activities (12, 21). While human and black flying fox IFIT1 share only 69% identity, human and chimpanzee IFIT1 are over 98%

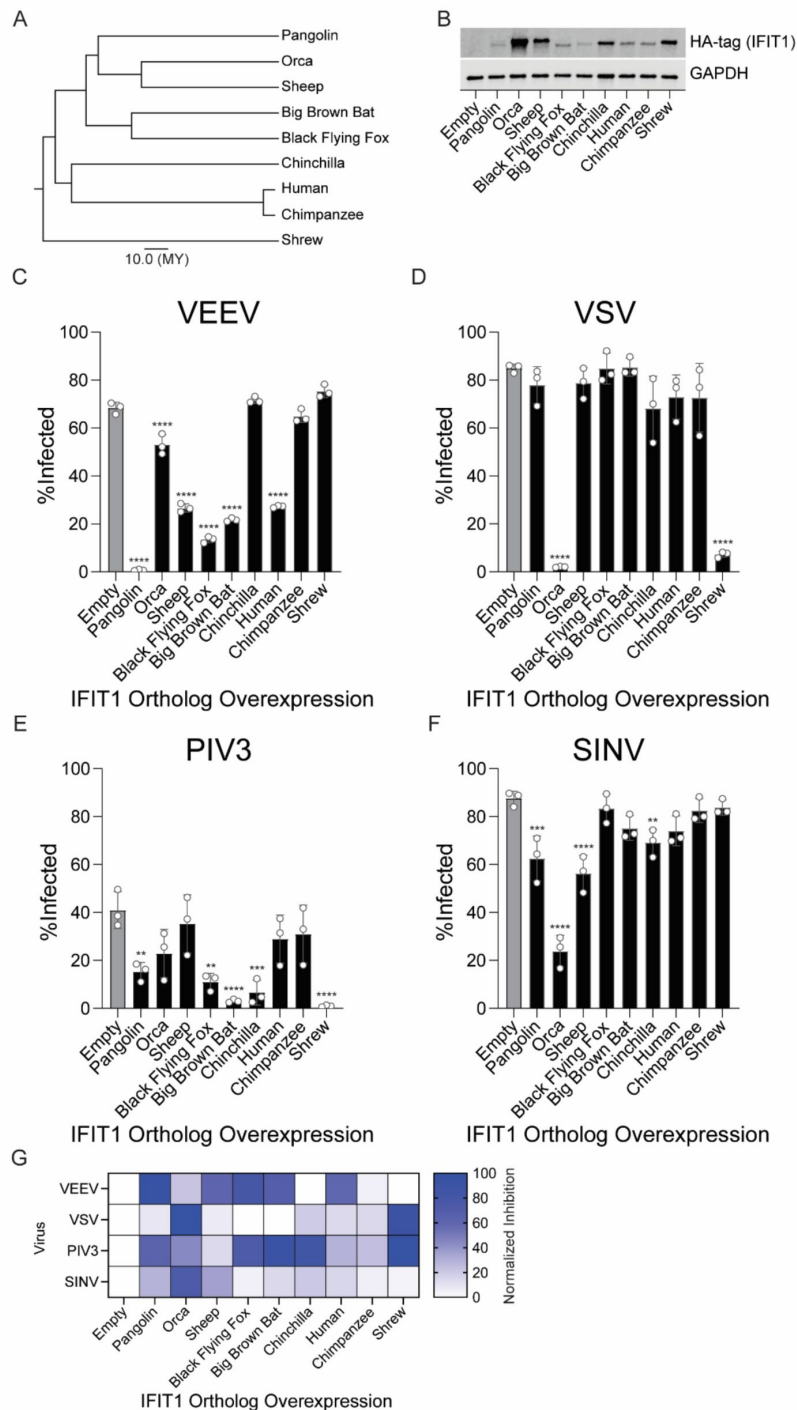


Figure 4.

Validation of selected IFIT1s from ortholog screen.

A. TimeTree illustrating the evolutionary relationship of IFIT1 orthologs selected for screen validation and follow-up. Scale bar represents divergence time of 10 million years. B. Western blot from cells expressing HA-tagged IFIT1 orthologs used in C-F. Image is representative of three independent replicates. C. Infection of IFIT1 ortholog-expressing Huh7.5 cells with VEEV-GFP (MOI 2, 4h) D. Infection of IFIT1 ortholog-expressing Huh7.5 cells with VSV-GFP (MOI 2, 4h) E. Infection of IFIT1 ortholog-expressing Huh7.5 cells with PIV3-GFP (MOI 2, 10h) F. Infection of IFIT1 ortholog-expressing Huh7.5 cells with SINV-GFP (MOI 2, 10h). In C-F, data represent mean $\pm$ SD, $n=3$ biological replicates. Statistical significance was determined by one-way ANOVA with Dunnett's test. ns $p > 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$.

identical (Fig 3B, Supp Table 1). Therefore, IFIT1 from these three species were of particular interest for functional studies. Accordingly, we sought to determine if this variability in restricting VEEV was due to differences in Cap0 RNA-binding ability of IFIT1 orthologs.

To perform *in vitro* RNA-binding assays, recombinant IFIT1 proteins were generated (Supp Fig 2). We then performed an electrophoretic mobility shift assay (EMSA) to investigate the RNA-binding properties of three mammalian IFIT1 orthologs. RNA probes matching the first 41 nucleotides of the VEEV 5' UTR were generated for EMSA by *in vitro* transcription. Probes were capped by the faustovirus capping enzyme (22) to generate 5' ends that resemble physiological Cap0 VEEV RNAs. After incubation of IFIT1 protein with Cap0 RNA probes, protein:RNA complexes were run on native gels, which were stained with a nucleic acid dye to visualize band shifts (Fig 5A). Quantification of band shift intensity and area under the curve analysis revealed that black flying fox IFIT1 exhibited the strongest RNA binding activity (Fig 5B, 5C). Human IFIT1 also bound VEEV RNA, but to a lesser extent than black flying fox IFIT1. Little binding was observed by chimpanzee IFIT1, as band shift intensity was barely above that of a human RNA-binding deficient control (human IFIT1-R187H) (2, 3) (Fig 5B, 5C). These biochemical data are consistent with viral phenotypes (Fig 4C), suggesting that species-specific antiviral activities of IFIT1s can be partly explained by RNA binding potential.

Mutagenesis of primate IFIT1 reveals genetic determinants of IFIT1 antiviral function

While human IFIT1 exhibited potent VEEV restriction and significant VEEV Cap0 RNA binding, chimpanzee IFIT1 lacked these properties. Chimpanzee and human IFIT1 differ by only 8 amino acids (Fig 6A), prompting us to examine which residues underlie the phenotypic differences. We performed site-directed mutagenesis to generate point mutations in both human and chimpanzee IFIT1 that correspond to the divergent residues between the two proteins (Fig 6A). As 3 of the 8 residues, 362, 364, and 366, are in close proximity (Fig 6A), we also generated a triple mutant IFIT1 that has these residues swapped between chimpanzee and human IFIT1 (hereafter referred to as the 362/4/6 triple mut). Notably, one of these three sites, L366 in humans and M366 in chimpanzee, was determined to be rapidly evolving in 2 of the tested computational models (Fig 1D).

We transduced Huh7.5 cells with lentiviruses expressing wild type (WT), single point mutant IFIT1s, or 362/4/6 triple mut IFIT1. After selection of transduced cells with puromycin, stable cells were challenged with VEEV-GFP at an MOI of 2 and infectivity was quantified by flow cytometry (Fig 6B). Western blotting demonstrated near uniform expression of all human and chimpanzee IFIT1 mutants (Fig 6B). Cells expressing wild-type chimpanzee IFIT1 were infected by VEEV at a rate of about 50%, similar to the approximately 60% infection rate observed in control cells. WT human IFIT1-expressing cells were infected at a rate of less than 10%, demonstrating significant VEEV suppression. Interestingly, the only point mutation in human IFIT1 that resulted in a significant loss of antiviral activity was L366M. Cells expressing human IFIT1 with the L366M point mutant were infected at a rate roughly 4-fold higher than cells expressing WT human IFIT1 (Fig 6B). Introduction of a Q364R mutation in human IFIT1 also resulted in a partial loss of antiviral function, although this was not statistically significant. Furthermore, the 362/4/6 triple mut introduced into human IFIT1 resulted in a near complete loss of anti-VEEV function of human IFIT1 (Fig 6B). The reciprocal was true for chimpanzee IFIT1, which significantly gained antiviral function when the M366L mutation was introduced. Cells expressing chimpanzee IFIT1 M366L were roughly half as likely to be infected as cells expressing the WT chimpanzee IFIT1 (Fig 6B). In addition, R364Q mutant chimpanzee IFIT1 resulted in a moderate increase in antiviral activity; however, this was not statistically significant compared to wild-type chimpanzee IFIT1. Furthermore, the 362/4/6 triple mut chimpanzee IFIT1 potently suppressed VEEV infection at a nearly identical level to that of WT human IFIT1. Together, we show that residues N362, Q364, and

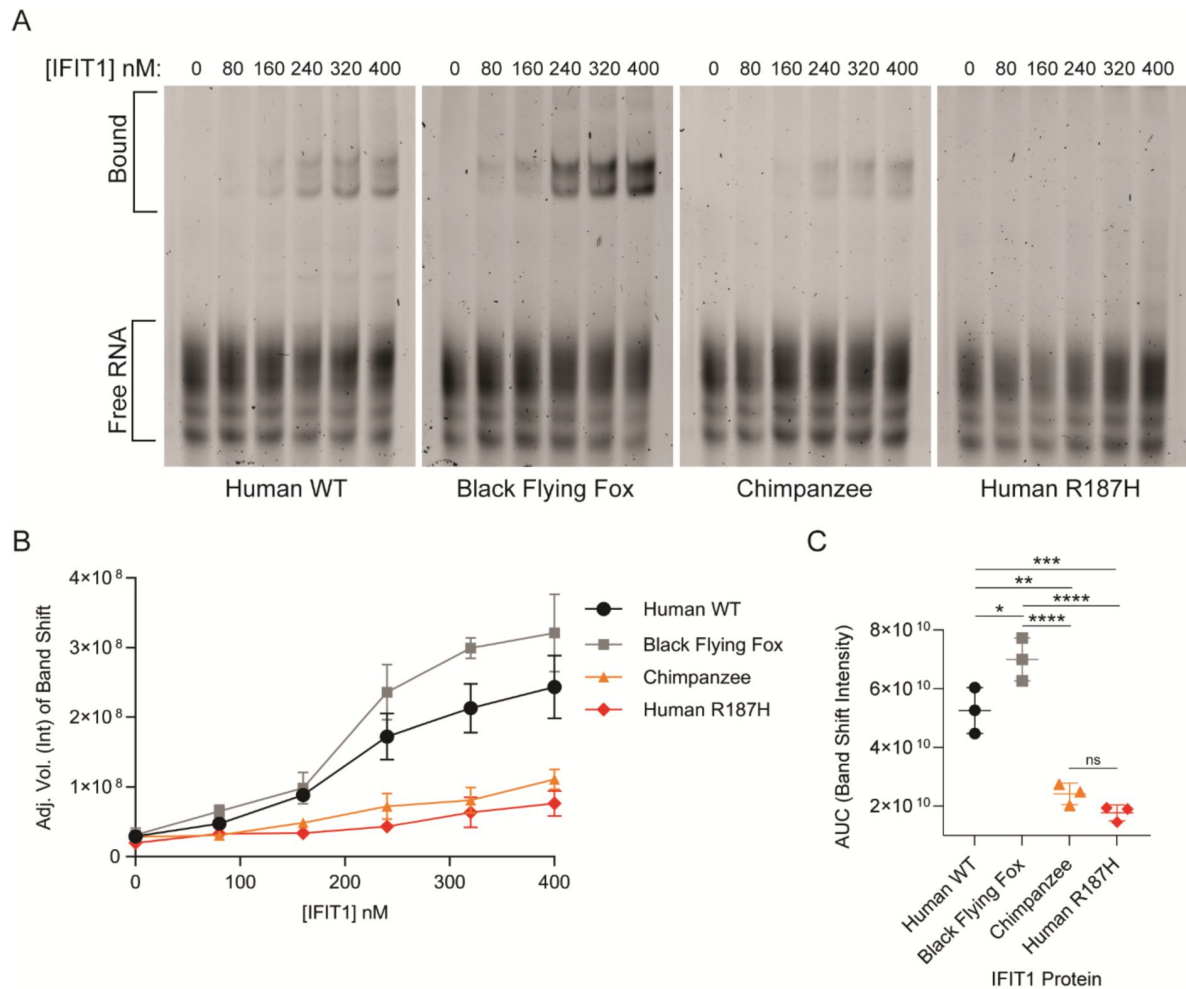


Figure 5.

IFIT1 proteins exhibit species-specific Cap0 RNA binding.

A. RNA EMSA with Cap0 VEEV RNA probes (41nt of VEEV TC-83 strain 5' UTR) at 50nM incubated with increasing concentrations of the indicated purified IFIT1 proteins. Images are one representative image from three independent replicates. B. Plot of the band shift intensity from EMSAs in A. Data represent mean $\pm$ SD, $n=3$. Band intensity was quantified by ImageLab software (BioRad). C. Area Under the Curve (AUC) analysis calculated from raw data in B. Statistical significance was tested by one-way ANOVA with correction for multiple comparisons. Data represent mean $\pm$ SD, $n=3$ replicates. ns, $p > 0.05$, *, $p < 0.05$, **, $p < 0.01$, ***, $p < 0.001$, and **** $p < 0.0001$.

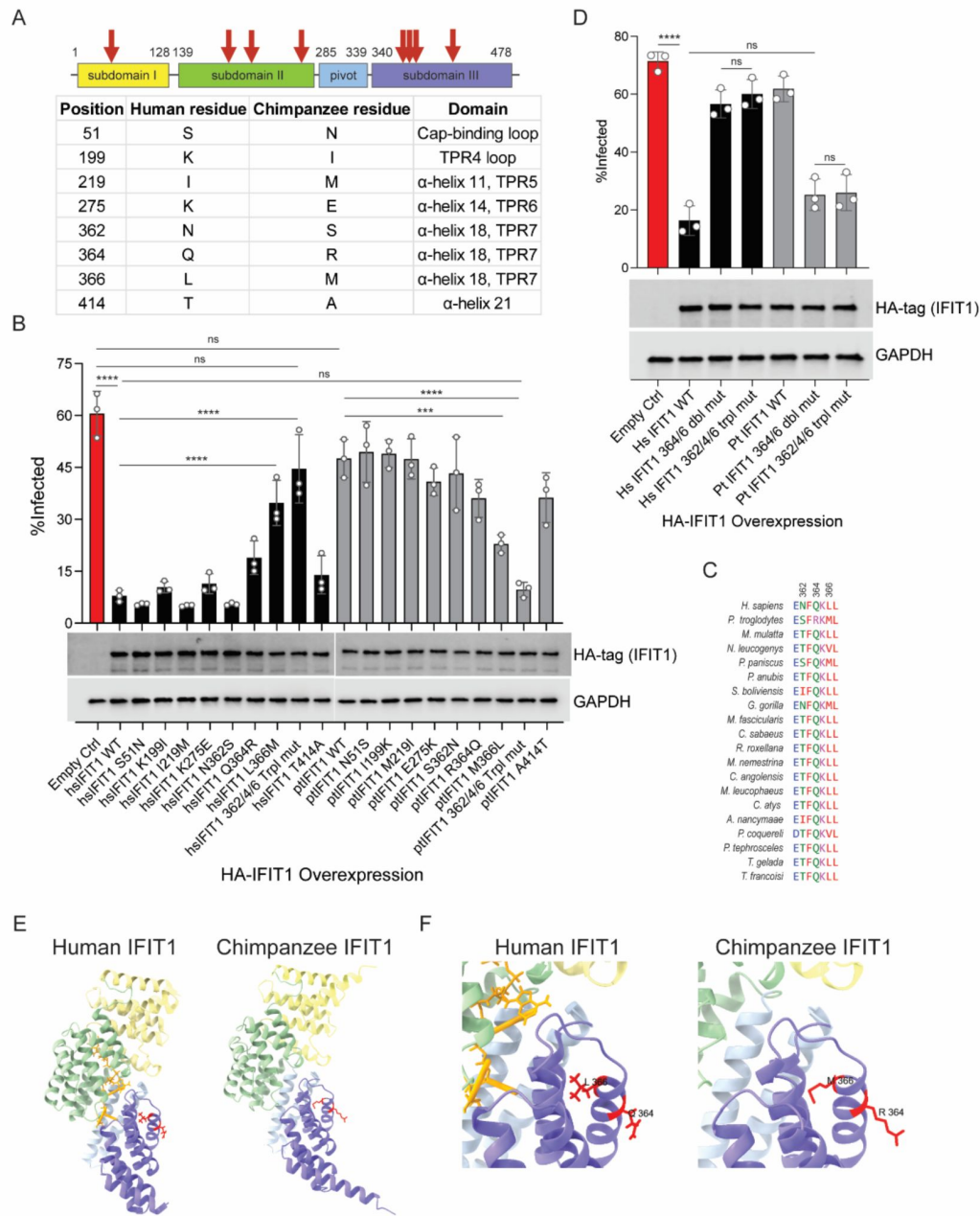


Figure 6.

Mutagenesis uncovers genetic determinants of primate IFIT1 antiviral function.

A. Diagram and chart describing amino acids that differ between human and chimpanzee IFIT1 as well as their domain location. B. Effects of primate IFIT1 mutant expression on VEEV-GFP infection. Huh7.5 cells were infected with an MOI of 2 for 4 h and infectivity was quantified by flow cytometry. Data represent mean $\pm$ SD, n=3 biological replicates. Statistical significance was determined by one-way ANOVA with Dunnett's test. ns, $p > 0.05$; ***, $p < 0.001$; and **** $p < 0.0001$. Western blotting for HA-tag (IFIT1) and GAPDH (Loading control) was also performed to determine expression levels of IFIT1 mutants. C. ClustalOmega protein sequence alignment of residues 361-367 of IFIT1 from 20 primate species. D. Effects of primate IFIT1 double or triple mutants on VEEV-GFP infection. Huh7.5 cells were infected with an MOI of 2 for 4 h and infectivity was quantified by flow cytometry. Data represent mean $\pm$ SD, n=3 biological replicates. Statistical significance was determined by one-way ANOVA with Dunnett's test. ns, $p > 0.05$; and **** $p < 0.0001$. Western blotting for HA-tag (IFIT1) and GAPDH (Loading control) was also performed to determine relative IFIT1 mutant protein abundance. E. Structure of human IFIT1 bound to RNA (left) or AlphaFold predicted structure of chimpanzee IFIT1 (right) visualizing location of residues 364 and 366 (red) that confer antiviral activity. Graphics were generated using ChimeraX. F. Zoom in of (E).

L366 in human IFIT1 are necessary for viral suppression of VEEV. In addition, the mutations of S362N, R364Q, and M366L in combination are sufficient for chimpanzee IFIT1 to gain potent antiviral function.

Amino acids 362, 364, and 366 are generally conserved across IFIT1 from 20 different primate species (**Fig 6C**). Specifically, residue 366 is typically a hydrophobic amino acid, with 5 out of the 20 primate sequences analyzed having either a valine or methionine, as seen in chimpanzee IFIT1, instead of a leucine, which is present in human IFIT1 (**Fig 6C**). This suggests that leucine at position 366 might be important for robust antiviral function in primate IFIT1, although further experiments are necessary to confirm this role. Additional support for the importance of L366 comes from our findings that the only primate IFIT1, besides those of chimpanzees and humans, which was expressed at appreciable levels during the screen was that of the Angola colobus. This IFIT1 variant also potently inhibited VEEV infection and notably contains a leucine at residue 366, similar to human IFIT1 (**Fig 3C**, **Supp Fig 1**). Glutamine is conserved at site 364 in all primate species aligned except chimpanzee which encodes an arginine (Q364R), introducing a positive charge. Point mutants of residue 364 in chimpanzee and human IFIT1 exhibited moderate alterations in anti-VEEV function, though not statistically significant, suggesting that this amino acid may have a minor role in antiviral restriction that synergizes with L366.

To test if residues 364 and 366 synergize to confer antiviral specificity between human and chimpanzee IFIT1, 364 and 366 double mutants (referred to as 364/6 double mut) were generated for both chimpanzee and human IFIT1. Huh7.5 cells expressing to express wild-type (WT), 364/6 double mut, or 362/4/6 triple mut IFIT1s of both chimpanzee and human IFIT1 were then challenged with VEEV-GFP at an MOI of 2 and infection was quantified by flow cytometry (**Fig 6D**). Western blotting demonstrated uniform expression of all human and chimpanzee IFIT1 mutants (**Fig 6D**). As in previous experiments, wild-type human IFIT1 strongly suppressed VEEV infection while wild-type chimpanzee IFIT1 did not (**Fig 6D**). However, the 364/6 double mut and 362/4/6 triple mut of human IFIT1 both had a near complete loss of antiviral capacity, resulting in cells being infected at a similar rate as cells expressing the wild-type chimpanzee IFIT1 (**Fig 6D**). These results suggest that residues 364 and 366, but not residue 362, in human IFIT1 are necessary to confer anti-VEEV function. In contrast, cells expressing the 364/6 double mutant or the 362/4/6 triple mutant of chimpanzee IFIT1 potently suppressed VEEV infection at rates similar to those observed in cells expressing wild-type human IFIT1. No difference in antiviral potency was observed between chimpanzee IFIT1 with the 364/6 double mut or 362/4/6 triple mut, suggesting again that residue 362 does not play a key role in the differential antiviral properties of human and chimpanzee IFIT1. Together, these data indicate that out of 8 amino acids that differ between human and chimpanzee IFIT1, sites 364 and 366, in combination, comprise the underlying differences in anti-VEEV function.

We next examined the location of amino acids 364 and 366 in the solved crystal structure of human IFIT1 bound to RNA and in the chimpanzee IFIT1 structure predicted by AlphaFold. Residues 364 and 366 are located within the α -helix 18 within TPR7 of IFIT1 (**Fig 6A, 6E**). They are not predicted to contact RNA and are not located in direct proximity of the RNA-binding pocket (**Fig 6E, 6F**), suggesting they may regulate IFIT1 function in an allosteric manner.

Discussion

Here, combining evolutionary and functional data, we conclude that mammalian IFIT1s are rapidly evolving viral restriction factors that exhibit species-specific antiviral activity. Furthermore, differences between the ability of human and chimpanzee IFIT1 to restrict VEEV led to the unexpected discovery that 2 residues in α -helices 18 within the TPR7 domain of IFIT1 may partially determine primate IFIT1 antiviral capacity. Swapping amino acids 364 and 366 of the IFIT1 protein sequence between human and chimpanzee IFIT1, in combination, resulted in

reciprocal antiviral phenotypes (**Fig 6B, 6D**). The structures of human and chimp IFIT1 suggest that these amino acids are not directly involved in RNA binding. Thus, they may be involved in conformational changes of IFIT1 during RNA-binding or other allosteric effects, though additional studies are needed to test this.

Our data also support a hypothesis in which the breadth of IFIT1 antiviral phenotypes may be, at least partially, explained by differences in RNA-binding ability. VEEV infection was most potently suppressed by black flying fox IFIT1, significantly inhibited by human IFIT1, and unaffected by chimpanzee IFIT1 (**Fig 4C**). This trend was similarly observed in RNA binding assays with a sequence identical to that of the Cap0 VEEV 5' UTR by EMSA (**Fig 5**). Black flying fox IFIT1 exhibited the strongest binding of VEEV Cap0 RNA, pronounced binding was observed by human IFIT1, with minimal VEEV Cap0 RNA binding by chimpanzee IFIT1 (**Fig 5**). These data are consistent with previous studies showing that the ability of IFIT1 to bind viral RNA is required for its antiviral function (4, 23).

The differential antiviral activity between IFIT1B and IFIT1 across several mammals was previously demonstrated by Daugherty et al. (11). This study used phylogenetic approaches to show that gene birth, gene loss and gene conversion occurred throughout vertebrate *IFIT* genes, resulting in extensive genetic diversity (11). Furthermore, through viral infection assays and a yeast growth assay, the authors suggest different RNA-binding properties and disparate antiviral functions between primate IFIT1s and mouse Ifit1b (11). Our studies significantly expand the diversity of mammalian IFIT1s interrogated for antiviral function. Notably, based on our studies, IFIT1s across *Mammalia* are not universally antiviral against VEEV and VSV. In fact, antiviral specificity of IFIT1 orthologs is possibly quite narrow and highly dependent on which virus is being evaluated.

We additionally provide new data to support a model in which mammalian IFIT1s spanning diverse orders are rapidly evolving, and that in primates, this is occurring at functionally relevant sites that are not directly implicated in RNA binding. Our computational studies also identified amino acid 193 as rapidly evolving and a mutational scanning determined that this residue is mutationally resilient. This was similarly observed for TRIM5α, in which most mutations within the v1 loop did not affect antiviral potency (24). In addition, alteration of charge within the v1 loop of TRIM5α played a significant role in determining the antiviral outcome of TRIM5α mutants (24). We similarly observed that introduction of negatively charged residues at position 193 within the TPR4 loop of human IFIT1 ablated antiviral activity (**Fig 2D**). Therefore, mutational resiliency within a rapidly evolving residue of primate IFIT1 suggests that alteration of a single amino acid within restriction factor IFIT1 may not result in major gain or loss antiviral function. This may maximize the potential for adaptation during evolutionary arms races with viruses. However, we cannot rule out the potential major role of mutations at position 193 in combination with other mutations in IFIT1. Together, our data suggest that positions 193 is rapidly evolving in primates and may accommodate many different amino acids while retaining antiviral function.

Our study does have several limitations, however. All IFIT1 ortholog screens were performed by expressing IFIT1 from distinct species in human cells, which may potentially affect natural antiviral function. For example, some IFIT1 orthologs may bind viral RNA, but not interface with human proteins to effectively suppress translation, resulting in no antiviral activity when expressed in human cells. Furthermore, our biochemical assays focus on IFIT1 binding to Cap0 RNA only. This is due to comparative studies that have demonstrated IFIT1 has the strongest affinity for Cap0 RNA when compared to other RNAs, such as ppp-RNA (3). This allowed us to use human IFIT1 as a positive control for Cap0 RNA-binding with a strong signal-to-noise for comparative RNA binding assays between IFIT1 orthologs that were successfully purified. However, it is possible that differences in ppp-RNA binding may underlie differences in the antiviral potency of IFIT1 towards viruses such as VSV and PIV3. This study also does not address the possibility that differential viral evasion of IFIT1 proteins contributes to species-specific

antiviral activity. Furthermore, we have only provided genetic, but not biochemical or structural evidence for underlying differences in chimpanzee and human IFIT1 RNA-binding and antiviral activity. By mutagenesis assays we demonstrate that 2 amino acid changes confer antiviral activity between chimpanzee and human IFIT1. Additional insight into the role of these residues might be gained from molecular dynamics simulations using the crystal structure of human IFIT1 bound to Cap0 RNA as a template. Comparisons with the 364/366 double mutant might determine whether these substitutions alter conformational dynamics in a manner that disfavors RNA binding.

Summarily, our findings underscore the notion that unbiased screening, investigations of restriction factors from non-model mammalian species, and evolutionary-guided approaches may uncover novel findings in host-pathogen interactions.

Materials and methods

Cells and cell culture

Huh7.5 and 293T cells (from C. Rice, The Rockefeller University) were maintained in “complete” DMEM (Gibco): DMEM supplemented with 10% FBS (Gibco) and 1× non-essential amino acids (NEAA; Gibco). All derived stable cells expressing selectable markers were grown in complete DMEM containing puromycin (Gibco; Huh7.5: 4μg/mL). All cells were incubated at 37°C in 5% CO₂. All cell lines have been STR profiled and confirmed negative for mycoplasma contamination by PCR testing (Venor GeM mycoplasma detection kit, Sigma).

Viruses and viral infections

Venezuelan equine encephalitis virus (TC-83 strain) infectious clone was obtained from I.Frolov. VEEV-GFP stocks were generated as previously described (25 [↗](#)). Briefly, infectious clone plasmid was linearized, and RNA was in vitro transcribed with the mMESSAGE mMACHINE™ SP6 Transcription Kit (Invitrogen) followed by RNA clean-up with the MEGAclean™ Transcription Clean-Up Kit (Invitrogen). BHK-21J cells (8 × 10⁶ cells) were then electroporated with 5μg of in vitro transcribed RNA and plated. Media was changed 18 h after plating, and the remaining supernatant was collected 8 h post-media change. After centrifugation to remove cell debris, virus-containing supernatants were aliquoted and stored at -80°C. Sindbis virus (SINV-GFP, clone S300 from M. Heise) (26 [↗](#)) was generated as previously described. Vesicular stomatitis virus (VSV-GFP) (from J. Rose) and human parainfluenza virus type 3 (PIV3-GFP) (JS strain, from Peter Collins) were propagated by passaging in BHK-21J cells and storing clarified supernatants as virus stocks at -80°C. All viral infections were performed in DMEM supplemented with 1% FBS and 1x non-essential amino acids at a total volume of 0.2mL in 24-well tissue culture plates. After a 1 h incubation period, 0.3mL of complete DMEM was added to all wells until harvest. For analysis of viral infection by flow cytometry, cells were dissociated from the plate with 150μL Accumax (Innovative Cell Technologies) and then fixed in paraformaldehyde at a final concentration of 1%. Cells were allowed to fix for at least 30 min at 4°C and subsequently centrifuged at 1500 × g for 5 min. Supernatant was removed from cell pellets and fixed cells were then resuspended in FACS buffer (PBS [Gibco] supplemented with 3% FBS). Flow cytometry was performed on the Stratifiedigm S1000EON benchtop flow cytometer, with analysis in FlowJo software.

Cloning and plasmids

IFIT1 coding sequences were obtained from the NCBI database. IFIT1 sequences were cloned into the Gateway expression vector pSCRPSY (27 [↗](#)). IFIT1 orthologs were synthesized as gBlocks (IDT) with Gateway compatible DNA sequences and an N-terminal HA tag. Human and chimpanzee IFIT1 point mutants were generated using site-directed mutagenesis PCR. Human and chimpanzee IFIT1 362/4/6 triple mutants and 364/6 double mutants were generated by Gibson cloning of

gBlocks (IDT). For protein purification, IFIT1 protein coding sequences that were codon-optimized for bacterial expression were cloned into a modified pET28a bacterial expression vector, containing an N-terminal 6X-His tag followed by the yeast Sumo (smt3).

Western Blotting

Cells were lysed with M-PER (Mammalian Protein Extraction Reagent, Thermo Scientific) containing 1x complete protease inhibitor cocktail (Roche) at 4°C for 5 min with intermittent rocking. Lysate was then stored at -80°C. For running SDS-PAGE, lysate was thawed on ice, and mixed to a final concentration of 1x SDS Loading Buffer (0.2 M Tris-Cl pH 6.8, 5% SDS, 25% Glycerol, 0.025% Bromophenol Blue, and 6.25% beta-mercaptoethanol). Samples were run on 12% TGX FastCast acrylamide gels (Bio-Rad) or 4-20% Mini-PROTEAN® TGX™ precast protein gels (BioRad) and transferred to nitrocellulose membranes using a TransBlot Turbo system (Bio-Rad). Membranes were blocked with 5% milk in 1x Tris Buffered Saline, with Tween-20 (TBS-T [20mM Tris, 150mM NaCl, 0.1% Tween-20]) for at least 45min. Subsequent incubation with primary antibody (at dilutions between 1:500 to 1:5000) diluted in 1% milk in TBS-T was performed overnight, rocking at 4°C. Primary antibodies probing for GAPDH (Proteintech Cat# 10494-1-AP, RRID:AB_2263076 or Proteintech Cat# 60004-1-Ig, RRID:AB_2107436), IFIT1 (Proteintech Cat# 23247-1-AP, RRID:AB_2811269) and HA-tagged IFIT1 (Biolegend Cat# 901502, RRID:AB_2565006) were used in this study. After incubation with primary antibodies, membranes were washed 3 times with TBS-T by rocking for 5 min. Secondary antibodies (Goat anti-Rabbit 800CW and Goat anti-Mouse 680RD [LI-COR] at 1:5000 dilutions) diluted in 1x TBS-T were incubated with membranes for at least 45 min. Membranes were then washed 3 times with TBS-T by rocking for 5 min. Proteins were visualized using the LI-COR Odyssey FC imaging system.

Lentivirus production and transduction

To generate lentivirus pseudoparticles, 350,000 293Ts were plated onto poly-lysine coated 6-well plates. Lentiviral packaging plasmids expressing VSV-g and Gag-pol were then mixed with SCRPSY plasmids at a ratio of 0.1µg:0.5 µg:2.5 µg, respectively. Plasmids were incubated for 20 min with a mix of XtremeGENE9 transfection reagent (Roche) and OptiMEM (Gibco). Transfection complexes were then added to 293T cells in 1mL of DMEM supplemented with 3% FBS dropwise. After 6 h of incubation, media was changed to 1.5 mL of DMEM supplemented with 3% FBS. Supernatants containing lentiviral pseudoparticles were harvested at 48 and 72 h post-transfection. Lentiviral pseudoparticles were then processed by adding 4µg/mL Polybrene (Sigma), 20mM HEPES, centrifuging at 1500rpm for 5min to remove cell debris, and storing at -80°C until transductions.

Lentivirus transductions were performed in 6-well plates after plating 350,000 cells (Huh7.5). Final volume of transductions was either 1mL or 1.5mL of lentivirus mixed with “pseudoparticle” DMEM (DMEM supplemented with 3% FBS, 4µg/mL Polybrene, and 20mM HEPES) After a 6 h incubation at 37°C, 2mL of complete DMEM was added to all transductions. Transduced cells were split into selection media 48h post-transduction and passaged for a minimum of three passages in selection media.

Protein Purification

IFIT1 protein coding sequences that were codon-optimized for bacterial expression were cloned into a modified pET28a bacterial expression vector, containing an N-terminal 6X-His tag followed by the yeast Sumo (smt3). Plasmids were transformed into Rosetta DE3 cells. Transformants were grown in Terrific Broth (TB) with 50 mg/L Kanamycin and 34 mg/L of Chloramphenicol and Antifoam B at 37 °C. At the OD₆₀₀ of 0.4, the temperature was lowered to 18 °C. Protein expression was induced at OD₆₀₀=3 by addition of IPTG to 0.4 mM final concentration. Cells were harvested after an overnight incubation by centrifugation (OD₆₀₀=~8.5) and lysed in the lysis buffer (50 mM Tris-HCl pH 8, 500 mM NaCl, 15 mM Imidazole, 1 mM PMSF, 4 mM β-ME) by sonication. Cell lysates were cleared by centrifugation at 35,000 x g for 30 minutes. The cleared lysate was applied onto a

gravity column with pre-washed Ni-NTA resin. Resin was washed with 50 mM Tris-HCl pH 8, 1 M NaCl, 30 mM Imidazole, 4 mM β -ME. Proteins were eluted with 50 mM Tris-HCl pH 8, 300 mM NaCl, 300 mM Imidazole, 2 mM DTT. Proteins were treated with Ulp1 protease overnight to remove the Sumo tag and passed over a size-exclusion column (HiLoad 16/600 Superdex 200 pg), equilibrated with 50 mM Tris 8.0, 150 mM NaCl, 1 mM DTT. Peak fractions were pooled, concentrated by ultrafiltration, and stored at -80°C until use. For SDS-PAGE gels of purified IFIT1 protein, 500nM of human, black flying fox, chimpanzee, and human R187H mutant IFIT1 were boiled in 4X Laemmli Sample Buffer (BioRad) with 1.25% BME and loaded into a 4-20% Mini-PROTEAN® TGX™ Precast Protein Gel (BioRad). Gels were run and subsequently incubated in Coomassie stain (40% ethanol, 10% glacial acetic acid, 0.1% Coomassie R-250) for 1 hour and destained overnight (10% ethanol, 7.5% glacial acetic acid). Gels were imaged with the ChemiDoc Imaging System (BioRad).

RNA Electrophoretic mobility shift assay

Protein stock concentration of IFIT1 orthologs was determined by diluting protein 1:5 in 1x TBE and measuring A280. Extinction coefficient was determined by ProtParam tool (ExPASy), and subsequently used to calculate and normalize protein concentrations between stocks.

Before loading, native gels (10% 19:1 Acrylamide: Bis-acrylamide, 25mM NaCl, 1x TBE, 0.0005% APS, 0.001% TEMED) were pre-run with 1x TBE in a cold room for at least 2.5 h. While gels were pre-running, binding reactions were set up at room temperature after thawing and keeping both RNA and protein stocks on ice. All dilutions of protein and RNA were in 1x reaction buffer (50mM TRIS pH 8.0, 100mM NaCl, 1mM EDTA, 0.01mg/mL Heparin sodium sulfate, 2.5% Glycerol). For binding reactions, 50nM of Cap0 RNA was mixed 1:1 with 0, 80, 160, 240, 320, and 400nM of purified IFIT1 protein in a final volume of 20 μ L and incubated at room temperature for 20 min.

18 μ L of RNA:Protein complexes were then run on pre-run native gels for 2.5 h at 100V in 1x TBE in a cold room. Gels were then stained with 1x SYBR Gold stain (ThermoFisher Scientific) diluted in 1x TBE for 20 min rocking at room temperature and imaged on a ChemiDoc Imaging System (BioRad) with UV transillumination. Band intensity was subsequently analyzed using ImageLab software (BioRad).

In vitro transcription and 5' capping of RNA probes

Uncapped RNA probes were generated using the T7 RiboMAX™ Express Large Scale RNA Production System (Promega). Manufacturer instructions were modified for generating short RNA probes. Briefly, 10 μ M of DNA oligos corresponding to the T7 ϕ 2.5 promoter sequence and the complementary T7 ϕ 2.5 promoter sequence followed by the first 41nt of the VEEV TC-83 strain 5' UTR sequence. Oligos were boiled at 95°C for 2-3 min and then slow cooled to room temperature to allow annealing. 1 μ M of the annealed DNA oligos served as the template per 20 μ L T7 transcription reaction. RNA was purified according to manufacturer instructions using phenol:chloroform nucleic acid extraction (Promega).

To generate Cap0 RNA, purified uncapped RNA probes were modified using the Faustovirus Capping Enzyme (FCE) (NEB) using modified manufacturer instructions. Briefly, 100 μ L reactions were composed of 20 μ L of uncapped RNA purified after T7 transcription (described above), 0.2mM SAM, 1mM GTP, 7500U of FCE, 1x FCE reaction buffer, 80U RNasin® Plus Ribonuclease Inhibitor (Promega), and nuclease-free water up to 100 μ L. Reactions were incubated at 42°C for 4.5 h and then another 0.5 h at 50°C . Cap0 RNA was then purified using the same phenol:chloroform nucleic acid extraction as performed after the T7 transcription reaction (Promega). Cap0 RNA was determined to be at least 95% Cap0 RNA (m7GpppA) by targeted LC/MS after Nuclease P1 (NEB) digestion at 37°C for 30 min followed by inactivation of enzyme at 95°C for 2 min. LC/MS was performed by the UT Southwestern Preclinical Pharmacology Core Facility.

Computational Evolutionary Analysis

IFIT1 coding sequences for all analyses were obtained from the NCBI database. Alignment of multiple sequences was performed using MUSCLE and ClustalW implemented in MEGA X (28). Maximum-likelihood trees were constructed in MEGA X, with standard settings. Trimmed alignments were then generated (29), and when required, ALTER(30) was used to convert file formats. MEME (31) and FUBAR (14) were performed with HyPhy (32) using the Datamonkey web application (33). To perform PAML analysis, we used default settings with the F3x4 codon frequency table within CodeML. Likelihood ratio tests were performed to compare model 7 versus model 8 and model 8 versus model 8a to determine the presence of positive selection. Bayes empirical Bayes testing was performed to determine specific residues that are adaptively evolving. Species trees were generated using the TimeTree database (34). FigTree software was then used to make trees shown in figures. Alignments and determination of amino acid identity to human IFIT1 was performed using ClustalOmega (35). Graphics comparing human and chimpanzee IFIT1 structures were generated using ChimeraX (36).

Statistical analysis

All statistical analyses were performed in GraphPad Prism 9. All graphs represent the mean of $n=3$ independent biological replicates with error bars representing standard deviation. ns, $p > 0.05$, *, $p < 0.05$, **, $p < 0.01$, ***, $p < 0.001$, and **** $p < 0.0001$.

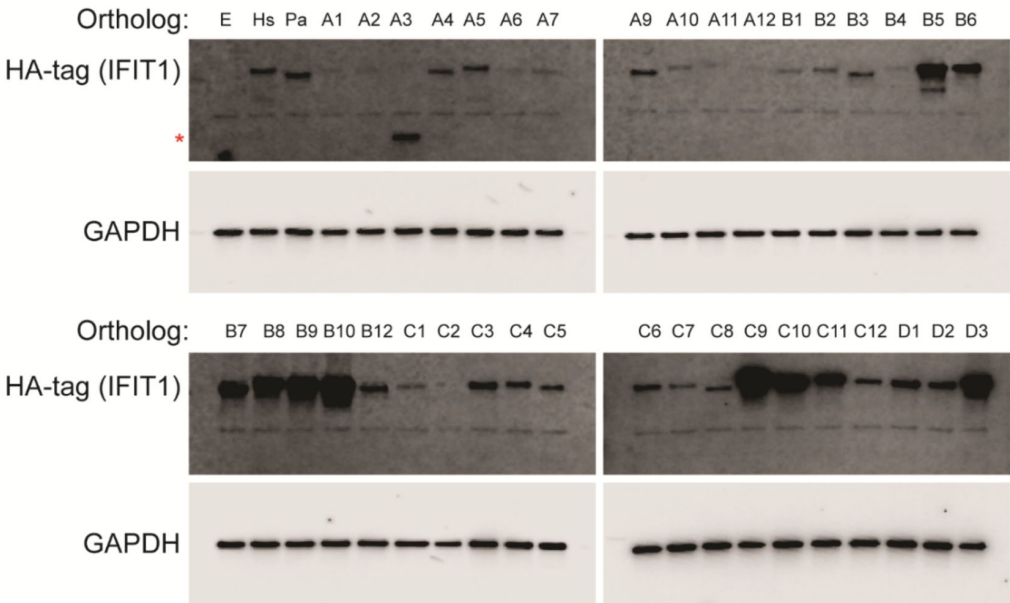
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Author Contributions

M.B.M, A.M.D., and E.N. performed technical experiments. M.B.M. and I.N.B. performed computational evolutionary analysis. M.B.M and J.W.S conceptualized this study and designed experiments. M.B.M and J.W.S wrote and edited the manuscript. M.B.M and J.W.S obtained funding.

A



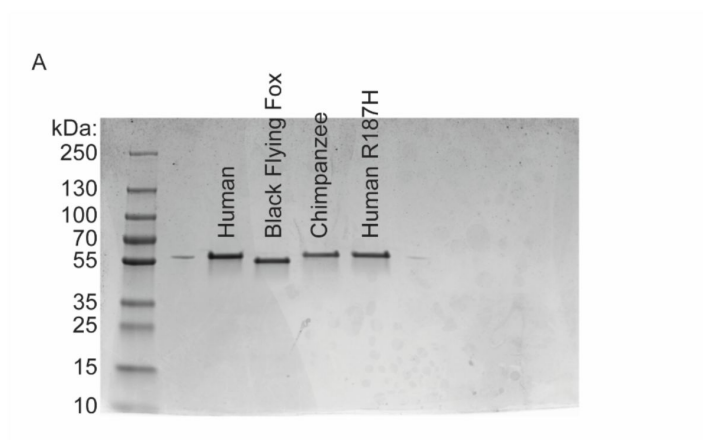
B

Common Name	Scientific Name	Key	Common Name	Scientific Name	Key
Empty Control	N/A	E	Cow	<i>Bos taurus</i>	B7
Human	<i>Homo sapiens</i>	Hs	Sheep	<i>Ovis aries</i>	B8
Black flying fox	<i>Pteropus alecto</i>	Pa	White-tailed deer	<i>Odocoileus virginianus texanus</i>	B9
African green monkey	<i>Chlorocebus sabaeus</i>	A1	Killer whale	<i>Orcinus orca</i>	B10
Coquerel's sifaka	<i>Propithecus coquereli</i>	A2	Common bottlenose dolphin	<i>Tursiops truncatus</i>	B12
Black-capped squirrel monkey	<i>Saimiri boliviensis</i>	A3	Ferret	<i>Mustela putorius furo</i>	C1
Angola colobus	<i>Colobus angolensis</i>	A4	Cat	<i>Felis catus</i>	C2
Chimpanzee	<i>Pan troglodytes</i>	A5	Dog	<i>Canis lupus familiaris</i>	C3
Crab-eating macaque	<i>Macaca fascicularis</i>	A6	Grizzly bear	<i>Ursus arctos horribilis</i>	C4
Rhesus macaque	<i>Macaca mulatta</i>	A7	Harbor (common) seal	<i>Phoca vitulina</i>	C5
Large flying fox	<i>Pteropus vampyrus</i>	A9	California sea lion	<i>Zalophus californianus</i>	C6
Velvety free tailed bat	<i>Molossus molossus</i>	A10	Nine-banded armadillo	<i>Dasypus novemcinctus</i>	C7
Greater horseshoe bat	<i>Rhinolophus ferrumequinum</i>	A11	Chinese pangolin	<i>Manis pentadactyla</i>	C8
Pale spear-nosed bat	<i>Phyllostomus discolor</i>	A12	European rabbit	<i>Oryctolagus cuniculus</i>	C9
Big brown bat	<i>Eptesicus fuscus</i>	B1	African savannah elephant	<i>Loxodonta africana</i>	C10
Kuhl's pipistrelle	<i>Pipistrellus kuhlii</i>	B2	Cape elephant shrew	<i>Elephantulus edwardii</i>	C11
Natal long-fingered bat	<i>Miniopterus natalensis</i>	B3	Linnaeus's two-toed sloth	<i>Choloepus didactylus</i>	C12
Horse 1	<i>Equus caballus</i>	B4	Ord's kangaroo rat	<i>Dipodomys ordii</i>	D1
Pig	<i>Sus scrofa</i>	B5	Chinchilla	<i>Chinchilla lanigera</i>	D2
Dromedary camel	<i>Camelus dromedarius</i>	B6	Horse 2	<i>Equus caballus</i>	D3

Supplementary Figure 1.

IFIT1 ortholog expression varied during the ortholog screen.

A. Western blot from lysates of cells expressing HA-tagged IFIT1 orthologs for screen in Fig 1A demonstrating unequal IFIT1 expression patterns. Images are representative of three independent replicates. B. Key denoting which symbols correspond to HA-tagged IFIT1 ortholog. Red asterisk marks the size of the truncated or cleaved protein product in lane A3.



Supplementary Figure 2.

Gel image of recombinant IFIT1 protein.

A. Coomassie stained SDS-PAGE gel of purified IFIT1 protein used for RNA EMSA.

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Editors

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Reviewer #1 (Public review):

Summary:

McDougal et al. aimed to characterize the antiviral activity of mammalian IFIT1 orthologs. They first performed three different evolutionary selection analyses within each major mammalian clade and identified some overlapping positive selection sites in IFIT1. They found that one site that is positively selected in primates is in the RNA-binding exit tunnel of IFIT1 and is tolerant of mutations to amino acids with similar biochemical properties. They then tested 9 diverse mammalian IFIT1 proteins against VEEV, VSV, PIV3, and SINV and found that each ortholog has distinct antiviral activities. Lastly, they compared human and chimpanzee IFIT1 and found that the determinant of their differential anti-VEEV activity may be partly attributed to their ability to bind Cap0 RNA.

Strengths:

The study is one of the first to test the antiviral activity of IFIT1 from diverse mammalian clades against VEEV, VSV, PIV3, and SINV. Cloning and expressing these 39 IFIT1 orthologs in addition to single and combinatorial mutants is not a trivial task. The positive connection between anti-VEEV activity and Cap0 RNA binding is interesting, suggesting that differences in RNA binding may explain differences in antiviral activity.

Weaknesses:

The evolutionary selection analyses yielded interesting results, but were not used to inform follow-up studies except for a positively selected site identified in primates. Since positive selection is one of the two major angles the authors proposed to investigate mammalian IFIT1 orthologs with, they should integrate the positive selection results with the rest of the paper more seamlessly, such as discussing the positive selection results and their implications, rather than just pointing out that positively selected sites were identified. The paper should elaborate on how the positive selection analyses PAML, FUBAR, and MEME complement one another to explain why the tests gave them different results. Interestingly, MEME which usually provides more sites did not identify site 193 in primates that was identified by both PAML and FUBAR. The authors should also provide the rationale for choosing to focus on the 3 sites identified in primates only. One of those sites, 193, was also found to be positively selected in bats, although the authors did not discuss or integrate that finding into the study. In Figure 1A, they also showed a $dN/dS < 1$ from PAML, which is confusing and would suggest negative selection instead of positive selection. Importantly, since the authors focused on the rapidly evolving site 193 in primates, they should test the IFIT1 orthologs against viruses that are known to infect primates to directly investigate the impact of the evolutionary arms race at this site on IFIT1 function.

Some of the data interpretation is not accurate. For example:

(1) Lines 232-234: "...western blot analysis revealed that the expression of IFIT1 orthologs was relatively uniform, except for the higher expression of orca IFIT1 and notably lower expression of pangolin IFIT1 (Figure 4B)." In fact, most of the orthologs are not expressed in a "relatively uniform" manner e.g. big brown bat vs. shrew are quite different.

(2) Line 245: "...mammalian IFIT1 species-specific differences in viral suppression are largely independent of expression differences." While it is true that there is no correlation between protein expression and antiviral activity in each species, the authors cannot definitively conclude that the species-specific differences are independent of expression differences. Since the orthologs are clearly not expressed in the same amounts, it is impossible to fully assess their true antiviral activity. At the very least, the authors should acknowledge that the protein expression can affect antiviral activity. They should also consider quantifying the IFIT1 protein bands and normalizing each to GAPDH for readers to better compare protein expression and antiviral activity. The same issue is in Line 267.

(3) Line 263: "SINV... was modestly suppressed by pangolin, sheep, and chinchilla IFIT1 (Figure 4E)..." The term "modestly suppressed" does not seem fitting if there is 60-70% infection in cells expressing pangolin and chinchilla IFIT1.

(4) The study can be significantly improved if the authors can find a thread to connect each piece of data together, so the readers can form a cohesive story about mammalian IFIT1.

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Reviewer #2 (Public review):

McDougal et al. describe the surprising finding that IFIT1 proteins from different mammalian species inhibit the replication of different viruses, indicating that the evolution of IFIT1 across mammals has resulted in host species-specific antiviral specificity. Before this work, research into the antiviral activity and specificity of IFIT1 had mostly focused on the human ortholog, which was described to inhibit viruses including vesicular stomatitis virus (VSV) and Venezuelan equine encephalitis virus (VEEV) but not other viruses including Sindbis virus (SINV) and parainfluenza virus type 3 (PIV3). In the current work, the authors first perform evolutionary analyses on IFIT1 genes across a wide range of mammalian species and

reveal that IFIT1 genes have evolved under positive selection in primates, bats, carnivores, and ungulates. Based on these data, they hypothesize that IFIT1 proteins from these diverse mammalian groups may show distinct antiviral specificities against a panel of viruses. By generating human cells that express IFIT1 proteins from different mammalian species, the authors show a wide range of antiviral activities of mammalian IFIT1s. Most strikingly, they find several IFIT1 proteins that have completely different antiviral specificities relative to human IFIT1, including IFIT1s that fail to inhibit VSV or VEEV, but strongly inhibit PIV3 or SINV. These results indicate that there is potential for IFIT1 to inhibit a much wider range of viruses than human IFIT1 inhibits. Electrophoretic mobility shift assays (EMSAs) suggest that some of these changes in antiviral specificity can be ascribed to changes in the direct binding of viral RNAs. Interestingly, they also find that chimpanzee IFIT1, which is >98% identical to human IFIT1, fails to inhibit any tested virus. Replacing three residues from chimpanzee IFIT1 with those from human IFIT1, one of which has evolved under positive selection in primates, restores activity to chimpanzee IFIT1. Together, these data reveal a vast diversity of IFIT1 antiviral specificity encoded by mammals, consistent with an IFIT1-virus evolutionary "arms race".

Overall, this is a very interesting and well-written manuscript that combines evolutionary and functional approaches to provide new insight into IFIT1 antiviral activity and species-specific antiviral immunity. The conclusion that IFIT1 genes in several mammalian lineages are evolving under positive selection is supported by the data, although there are some important analyses that need to be done to remove any confounding effects from gene recombination that has previously been described between IFIT1 and its paralog IFIT1B. The virology results, which convincingly show that IFIT1s from different species have distinct antiviral specificity, are the most surprising and exciting part of the paper. As such, this paper will be interesting for researchers studying mechanisms of innate antiviral immunity, as well as those interested in species-specific antiviral immunity. Moreover, it may prompt others to test a wide range of orthologs of antiviral factors beyond those from humans or mice, which could further the concept of host-specific innate antiviral specificity. Additional areas for improvement, which are mostly to clarify the presentation of data and conclusions, are described below.

Strengths:

- (1) This paper is a very strong demonstration of the concept that orthologous innate immune proteins can evolve distinct antiviral specificities. Specifically, the authors show that IFIT1 proteins from different mammalian species are able to inhibit the replication of distinct groups of viruses, which is most clearly illustrated in Figure 4G. This is an unexpected finding, as the mechanism by which IFIT1 inhibits viral replication was assumed to be similar across orthologs. While the molecular basis for these differences remains unresolved, this is a clear indication that IFIT1 evolution functionally impacts host-specific antiviral immunity and that IFIT1 has the potential to inhibit a much wider range of viruses than previously described.
- (2) By revealing these differences in antiviral specificity across IFIT1 orthologs, the authors highlight the importance of sampling antiviral proteins from different mammalian species to understand what functions are conserved and what functions are lineage- or species-specific. These results might therefore prompt similar investigations with other antiviral proteins, which could reveal a previously undiscovered diversity of specificities for other antiviral immunity proteins.
- (3) The authors also surprisingly reveal that chimpanzee IFIT1 shows no antiviral activity against any tested virus despite only differing from human IFIT1 by eight amino acids. By mapping this loss of function to three residues on one helix of the protein, the authors shed new light on a region of the protein with no previously known function.

(4) Combined with evolutionary analyses that indicate that IFIT1 genes are evolving under positive selection in several mammalian groups, these functional data indicate that IFIT1 is engaged in an evolutionary "arms race" with viruses, which results in distinct antiviral specificities of IFIT1 proteins from different species.

Weaknesses:

(1) The evolutionary analyses the authors perform appear to indicate that IFIT1 genes in several mammalian groups have evolved under positive selection. However, IFIT1 has previously been shown to have undergone recurrent instances of recombination with the paralogous IFIT1B, which can confound positive selection analyses such as the ones the authors perform. The authors should analyze their alignments for evidence of recombination using a tool such as GARD (in the same HyPhy package along with MEME and FUBAR). Detection of recombination in these alignments would invalidate their positive selection inferences, in which case the authors need to either analyze individual non-recombining domains or limit the number of species to those that are not undergoing recombination. While it is likely that these analyses will still reveal a signature of positive selection, this step is necessary to ensure that the signatures of selection and sites of positive selection are accurate.

(2) The choice of IFIT1 homologs chosen for study needs to be described in more detail. Many mammalian species encode IFIT1 and IFIT1B proteins, which have been shown to have different antiviral specificity, and the evolutionary relationship between IFIT1 and IFIT1B paralogs is complicated by recombination. As such, the assertion that the proteins studied in this manuscript are IFIT1 orthologs requires additional support than the percent identity plot shown in Figure 3B.

(3) Some of the results and discussion text could be more focused on the model of evolution-driven changes in IFIT1 specificity. In particular, the chimpanzee data are interesting, but it would appear that this protein has lost all antiviral function, rather than changing its antiviral specificity like some other examples in this paper. As such, the connection between the functional mapping of individual residues with the positive selection analysis is somewhat confusing. It would be more clear to discuss this as a natural loss of function of this IFIT1, which has occurred elsewhere repeatedly across the mammalian tree.

(4) In other places in the manuscript, the strength of the differences in antiviral specificity could be highlighted to a greater degree. Specifically, the text describes a number of interesting examples of differences in inhibition of VSV versus VEEV from Figure 3C and 3D, but it is difficult for a reader to assess this as most of the dots are unlabeled and the primary data are not uploaded. A few potential suggestions would be to have a table of each ortholog with % infection by VSV and % infection by VEEV. Another possibility would be to plot these data as an XY scatter plot. This would highlight any species that deviate from the expected linear relationship between the inhibition of these two viruses, which would provide a larger panel of interesting IFIT1 antiviral specificities than the smaller number of species shown in Figure 4.

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Reviewer #3 (Public review):

Summary:

This manuscript by McDougal et al, demonstrates species-specific activities of diverse IFIT1 orthologs and seeks to utilize evolutionary analysis to identify key amino acids under positive selection that contribute to the antiviral activity of this host factor. While the authors identify

amino acid residues as important for the antiviral activity of some orthologs and propose a possible mechanism by which these residues may function, the significance or applicability of these findings to other orthologs is unclear. However, the subject matter is of interest to the field, and these findings could be significantly strengthened with additional data.

Strengths:

Assessment of multiple IFIT1 orthologs shows the wide variety of antiviral activity of IFIT1, and identification of residues outside of the known RNA binding pocket in the protein suggests additional novel mechanisms that may regulate IFIT1 activity.

Weaknesses:

Consideration of alternative hypotheses that might explain the variable and seemingly inconsistent antiviral activity of IFIT1 orthologs was not really considered. For example, studies show that IFIT1 activity may be regulated by interaction with other IFIT proteins but was not assessed in this study.

Given that there appears to be very little overlap observed in orthologs that inhibited the viruses tested, it's possible that other amino acids may be key drivers of antiviral activity in these other orthologs. Thus, it's difficult to conclude whether the findings that residues 362/4/6 are important for IFIT1 activity can be broadly applied to other orthologs, or whether these are unique to human and chimpanzee IFIT1. Similarly, while the hypothesis that these residues impact IFIT1 activity in an allosteric manner is an attractive one, there is no data to support this.

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Author response:

We are grateful to the reviewers for their thoughtful and constructive feedback on our manuscript. Based on the Public Reviews, we will address the concerns raised by each reviewer through a combination of new analyses, clarifications, and expanded discussion as outlined below:

Reviewer #1:

(1) Integration of Positive Selection Results:

We will enhance the integration of positive selection analyses throughout the manuscript. Specifically, we will discuss how the positively selected sites in primates, including site 193, inform IFIT1 function. We will expand the discussion to explain how PAML, FUBAR, and MEME complement each other and why MEME did not detect site 193 in primates. Additionally, we will provide a rationale for focusing on the three sites identified in primates and address the overlap with bat orthologs.

(2) Expression Levels and Antiviral Activity:

We acknowledge the variability in IFIT1 ortholog expression levels. To address this, we will quantify and normalize protein expression to GAPDH across all orthologs, allowing for a more accurate comparison of antiviral activity. We will revise the text to clarify that species-specific differences in viral suppression may be influenced by expression levels.

(3) Clarification of Terminology and Data Interpretation:

We will refine our description of the antiviral effects observed for SINV in Figure 4E. We will also revise statements related to protein expression in the relevant sections to improve accuracy.

| (4) *Cohesion of Data:*

We will work to more tightly connect the evolutionary analysis with the functional virology data, framing the manuscript around how positive selection shapes IFIT1 function across species.

| **Reviewer #2:**

| (1) *Recombination Analysis of IFIT1:*

We will conduct a recombination analysis using GARD from the HyPhy package to ensure that the signatures of positive selection are not confounded by recombination between IFIT1 and IFIT1B.

| (2) *Clarification of IFIT1 Homologs Studied:*

We will provide additional details on how IFIT1 orthologs were selected, including addressing the relationship between IFIT1 and IFIT1B. We will support this by presenting additional sequence comparisons to demonstrate the orthology of the proteins studied.

| (3) *Chimpanzee IFIT1 Loss of Function:*

We will revise the discussion of chimpanzee IFIT1 to better reflect the data.

| (4) *Presentation of Antiviral Specificity Data:*

We will include a supplementary table listing the percentage of infection normalized to control by VSV and VEEV for each ortholog to allow for clearer comparisons.

Additionally, we will provide an alternative visualization to better compare the data sets.

| **Reviewer #3:**

| (1) *Alternative Hypotheses for IFIT1 Antiviral Activity such as IFIT1-IFIT interactions:*

We will expand the discussion to consider alternative hypotheses, including the potential for IFIT1 activity to be regulated through interactions with other IFIT family members. Therefore, we will address how IFIT1-IFIT interactions may be broadly applicable to our findings with IFIT1 orthologs. In addition, we will clarify that we do not conclude that residues 362/4/6 are the sole drivers of antiviral specificity across the orthologs tested in this study.

| (2) *Generalization of Findings Across Orthologs:*

We acknowledge that the functional importance of residues 362/4/6 may not be generalizable across all orthologs. We will discuss this limitation more explicitly in the manuscript, while also expanding on how these findings apply specifically to primate IFIT1 orthologs.

We believe that these revisions will address the key concerns raised by the reviewers and strengthen the manuscript. We look forward to submitting the revised version for further consideration.

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